

arc_temp_humid - outstanding work

Updated 6 September 2026.

Everything described below is **not yet done**. What *is* done is in `CHANGELOG.md` ; the architecture is in `CLAUDE.md` .

1. Rename the GitHub repository - DONE

`actionresearchprojects/arc_tz_temp_humid` is now `arc_temp_humid` , and everything that depended on the old name has been dealt with.

The hazard here was that the main site's `sync-embedded.yml` derives its destination folder from the dispatch payload's `source_repo` . A rename alone would have started filling a new `embedded/arc_temp_humid/` while the published page went on iframing the old folder, frozen: no 404, no error, just a dashboard that silently stopped updating.

That was headed off by mapping **both** names to the same folder:

```
arc_tz_temp_humid|arc_temp_humid)    FOLDER="arc_temp_humid" ;;
```

so the repository and the site never had to change in lockstep. The live state:

URL	Status
<code>/graphs/arc-temp-humid/</code>	live, iframes <code>embedded/arc_temp_humid/</code>
<code>/graphs/arc-temp-humid/config</code>	live
<code>/explainers/arc-temp-humid/</code>	live, regenerated from the current README
<code>/graphs/arc-tz-temp-humid/</code>	redirects to the new address
<code>/explainers/arc-tz-temp-humid/</code>	redirects to the new address

The git remote, both workflow payloads and the header info-icon link now all use the new name.

The superseded `embedded/arc_tz_temp_humid/` copy has been removed from the site repo, and the data flow explainer regenerated so it documents the new name.

Note the Pages URL for this repo changed with the rename, from `actionresearchprojects.net/arc_tz_temp_humid/` to `.../arc_temp_humid/`. Nothing links to it - the public route is `/graphs/arc-temp-humid/`.

2. Automate the ARC UK Omnisense fetch

This is the only item still blocked on information I do not have.

The UK export is added **by hand** to

`data/omnisense_uk/omnisense_uk_YYYYMMDD_HHMM.csv`. The timestamp is the fetch time and is what the sidebar's freshness note reads, so keep the format.

`fetch_omnisense.py` is already parameterised by site. To switch it on:

1. Fill in the commented-out `SITES["uk"]` entry, in particular `site_nbr`.
2. Add a `--site uk` step to `.github/workflows/update-dashboard-data.yml`, next to the existing Omnisense step, with the same `continue-on-error`.

Finding the site number

It is not in the CSV: the export carries site *names*, not ids, and the `download_935909450.csv` filename was a download-job id.

Log in at omnisense.com, select the site, and read `siteNbr=` from the address bar. That is where the Tanzanian `152865` came from - it appears in `fetch_omnisense.py` as `dnld_rqst.asp?siteNbr=152865`.

The site is named "**Simmonds.Mills Retrofits**" - the `site_name` on every block of the UK export.

Credentials are probably already correct

The UK export contains a sensor labelled `House 5 Metal Roof S.E` alongside the Grove, Holywell, Chestnuts and No. 59 sensors, so one login already sees both the Tanzanian and UK material. The existing `OMNISENSE_USERNAME` / `OMNISENSE_PASSWORD` secrets will very likely work, with only `site_nbr` differing.

If not, `SITES["uk"]` accepts `username_env` / `password_env` naming its own secrets, falling back to the shared pair when absent.

When it runs

The UK export is one CSV covering several sites - Grove, Holywell, Chestnuts, No. 59 - of which this dashboard uses four sensors. `GROVE_SENSORS` and `HOLYWELL_SENSORS` in `build.py` filter by sensor id, so the rest are ignored rather than plotted. Those two sets are what to update if sensors change.

3. Decide the UK overheating threshold

Currently **none**: the three UK datasets have `"threshold": None`, so no red band is drawn and the sidebar control is hidden rather than offering a meaningless toggle. Tanzania keeps 32-35 C.

No UK figure was assumed because the sensible source, CIBSE TM52 / TM59, is **per room type**. TM59 uses 26 C for bedrooms, but Grove's sensor is a living room and Holywell's is "Internal Ambient".

Once decided it is one line per dataset:

```
"threshold": {"low": 26, "high": 28},
```

The label, the band on all three chart types and the sidebar control follow automatically.

4. Confirm the UK Open-Meteo coordinates

Done and live, but with one assumption worth correcting.

Both UK buildings now have their own Open-Meteo feed. Each drives its own building's adaptive comfort running mean and both appear on the ARC UK chart. They are fetched daily alongside the Tanzanian feed.

The coordinates in `fetch_openmeteo.py` are **town-centre positions** for Hereford and Criccieth, not the buildings themselves:

```
"grove": {"lat": 52.0567, "lon": -2.7160, ...} # Hereford
"holwell": {"lat": 52.9186, "lon": -4.2372, ...} # Criccieth
```

Open-Meteo's historical model runs on a roughly 11 km grid, so a town-centre fix is usually the same grid cell as the building - but if the exact positions (or postcodes) are to hand, replace the two lat/lon pairs. It is a one-line change per site and needs no other edits; the next daily run picks it up.

As a sanity check that the two feeds are genuinely distinct: Grove averages 1.03 C warmer than Holywell across 30,497 paired hours, with only 611 hours identical - consistent with inland Herefordshire against coastal North Wales.

5. Known limitation, not a task

ASHRAE 55 Section 5.4.1(a) requires that no heating is in operation. Nothing at any site records heating status, and nothing in a temperature and humidity trace reliably separates a heated room from a merely warm one.

The dashboard states this rather than inferring it: a bolded line in the sidebar applicability panel directly beneath the quoted criterion, and in the tooltips explaining the method. The 10-33.5 C running-mean gate is enforced automatically and greys out readings the standard does not cover.

That gate now has teeth it did not have before, because the UK Open-Meteo feeds reach back to March 2023: once UK winter indoor data exists, cold-weather readings will be greyed and excluded from the comfort percentage. But **a mild day with the heating on will still pass every test the dashboard can apply.**

Closing this properly needs a heating-status input, not a cleverer algorithm. See `adaptivecomfort.md` section 2 for the full reasoning, including why calendar-month filtering was considered and rejected.